

Vincent Tong

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RELEVANT EXPERIENCE

[Rate Pulse](#)

Perth, Australia

Software Engineer

December 2025 - Present

- Built and launched Rate Pulse, a production FX monitoring platform that consolidates fragmented rate-checking workflows with centralised exchange-rate comparison, conversion, historical trends, and macro-news context.
- Designed a secure, read-heavy backend for public FX browsing, user preferences, and admin workflows, using **Go/Gin**, **PostgreSQL/sqlc**, **Redis** caching, **Paseto/JWT** auth, Brevo-powered email verification, **REST APIs**, and **gRPC/Protocol Buffers** for typed service contracts.
- Developed a **Next.js / React / TypeScript** dashboard that helps users move from rate discovery to market context, combining source-aware rate tables, converter flows, historical Recharts visualizations, and macro-news feeds.
- Built **Python Selenium** pipelines to **collect and normalize** multi-source FX and market-intelligence data into **PostgreSQL** and market-intelligence snapshots into **MongoDB**, continuously refreshing dashboard tables, charts, and news feeds.
- Automated production delivery with **Docker**, **GitHub Actions**, **Kubernetes**, and **DigitalOcean**, reducing manual deployment work through independent API/worker rollouts and scheduled FX/news collection workflows.

[Lions Eye Institute](#)

Perth, Australia

Software Engineer Intern

July 2025 - November 2025

- Built a cross-platform medical application using **C++/Qt**, **Python** with a **SQLite** data layer, centralising ophthalmic instrument data and reducing inspection time by **89%**.
- Delivered an internal project management platform using **Next.js**, **ASP.NET Core**, **PostgreSQL**, centralising laboratory resources and schedules while reducing team context-switching.
- Developed an image-processing pipeline using a pretrained segmentation model, with modular preprocessing and evaluation scripts that achieved **83% accuracy** for retinal vessel segmentation across heterogeneous datasets.

[Samsung Corporation](#)

Bac Ninh, Vietnam

Software Engineer Intern

August 2023 - October 2023

- Developed a **C#/.NET (WPF)** desktop application with a **SQL Server** backend to centralize resource tracking, replacing fragmented spreadsheet workflows and improving cross-team operational visibility.
- Deployed a pretrained **computer vision** system for display panel defect detection, reducing manual inspection latency by **40%** and standardising production QA benchmarks.

[Vietnam Posts and Telecommunications Group](#)

Hanoi, Vietnam

Data Engineer

October 2022 - June 2023

- Built distributed data pipelines using **PySpark** and **Hadoop** to process **100M+ telecom records**, improving pipeline performance by **2x** and supporting downstream backend services.
- Engineered a full-stack recommendation engine using **FastAPI**, **Next.js**, **MongoDB**, and **Selenium** pipelines to surface real-time alternatives for out-of-stock products, reducing dead-end user flows.

CERTIFICATION

AWS Certified Developer – Associate (Valid until Feb 2029) - [\[Certificate Link\]](#)

PROJECTS

Book Borrow (August 2025 – November 2025) (Team of 8) | [GitHub](#)

- Engineered and delivered a production-ready P2P marketplace for a non-profit book reader organisation in Western Australia, using **FastAPI** and **Next.js** with integrated **Stripe** payments, **Google OAuth**, and **AWS RDS**.

TECHNICAL SKILLS

- Programming Languages:** Golang, Python, C/C++, JavaScript/TypeScript, Java, SQL.
- Frameworks & Libraries:** ASP.NET Core, Next.js, React, FastAPI, Gin, Flask, Qt, PySpark.
- Cloud & DevOps:** DigitalOcean, AWS, Docker, Kubernetes, GitHub Actions, Vercel, Render, Supabase, Upstash.
- Databases & Caching:** PostgreSQL, MySQL, MongoDB, Redis.
- Tools & Architectures:** Git, gRPC, RESTful APIs, Brevo, Selenium, Linux, Unit Testing, CI/CD Pipelines.

EDUCATION

[The University of Western Australia](#)

Perth, Australia

Master of Information Technology, Software Systems Specialisation

February 2024 – December 2025

[Hanoi University of Science and Technology](#)

Hanoi, Vietnam

Bachelor of Information Technology

August 2019 - September 2023